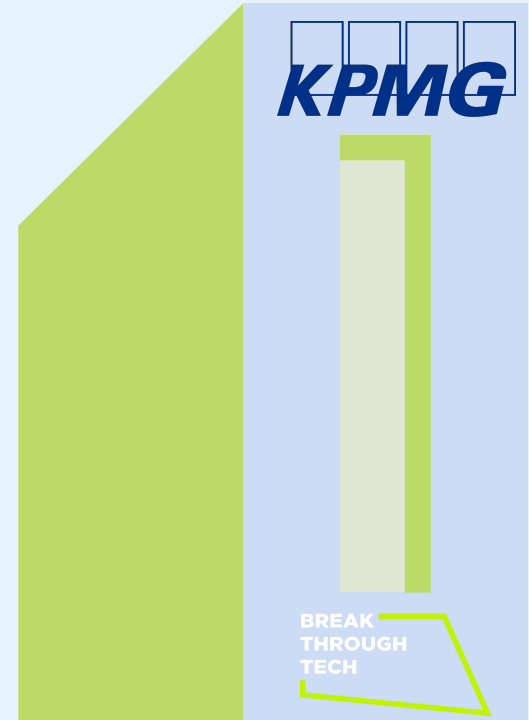


Measuring the Impact of AI Productivity, Sustainability, and Energy Optimization: Building Energy-AI Optimization

KPMG 1A



AGENDA

01

Introduction

02

Project Overview

03

Data Preparation Steps

04

Data Understanding & Model Evaluation

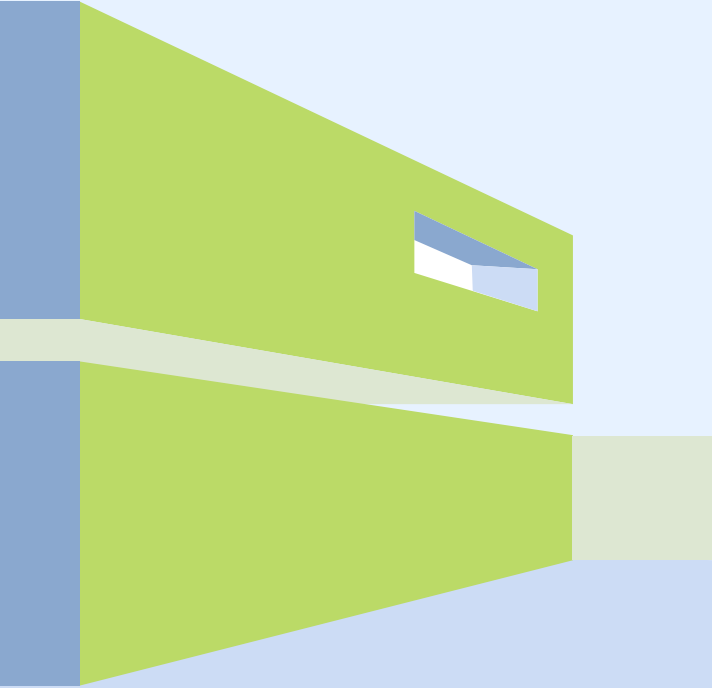
05

Conclusions + Next Steps

06

Questions





01

Introductions

MEET THE TEAM



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Stony Brook University



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City College of New York



Anveetha Suresh

The University of Texas at Dallas



Noble Adike

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AI Studio Coach



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Challenge Advisor



Reed Rozelle

Challenge Advisor



Laurence Brown

Challenge Advisor

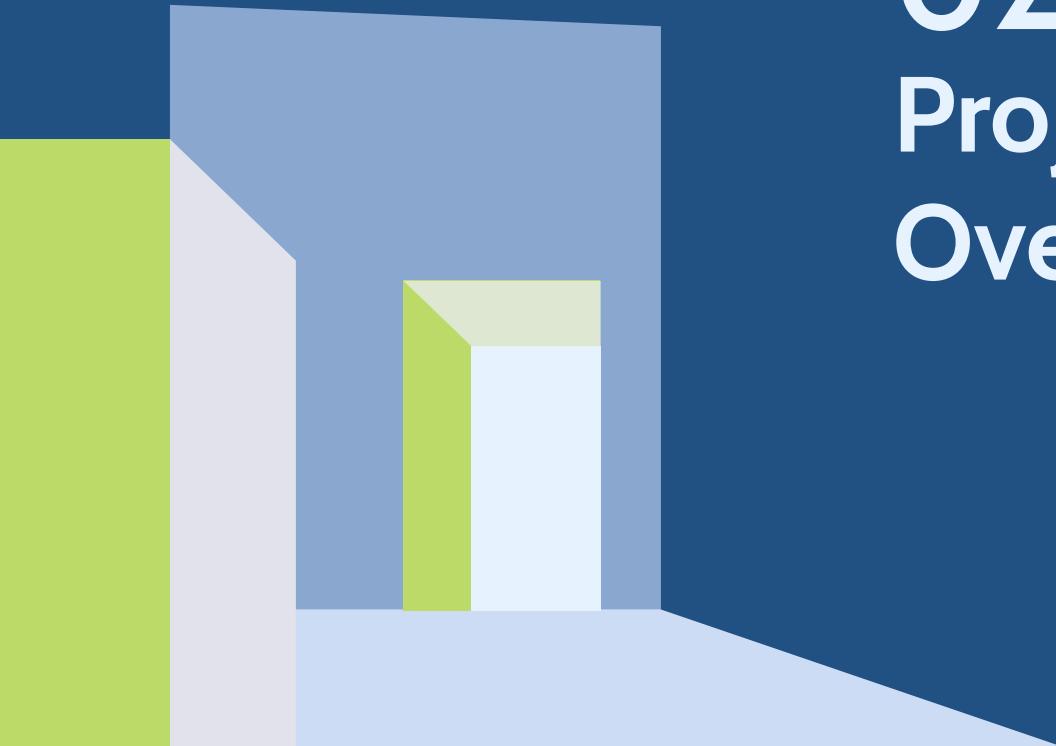


Chengwei Wang

Challenge Advisor



BREAK
THROUGH
TECH



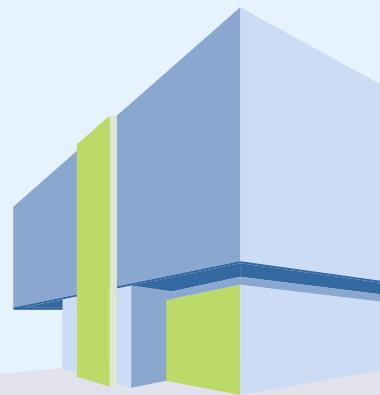
02 Project Overview

OUR MISSION

According to the U.S. Environmental Protection Agency (EPA), 30% of the energy used in commercial buildings is wasted.

We reframed that loss as a design opportunity by analyzing NREL Large Office Buildings time-series data to identify which end uses most drive site energy reductions.

Then we translated those findings into actionable recommendations for KPMG's new Manhattan workspace.



DATA OVERVIEW

- **Data Source:** NREL Large Office Buildings (time-series CSV)
- **Key Features:** IT/computing load, temp, humidity, lighting, total/site energy

Scope: Focus on drivers of **site energy reduction** relevant to office design/operations

OUR PROCESS

01

Data

Preparation

Collected and cleaned datasets (NREL).
Handled missing values, outliers, and aligned timestamps.

Exploratory Data Analysis (EDA)

Visualized energy and IT load patterns. Identified correlations and key predictive features.

02

03

Model

Development

Supervised regression time-series forecasting (**LSTM**) and evaluate (train/val/test + error metrics like MAE/RMSE)

Optimization & Insights

Tuned models for higher accuracy. Generated actionable recommendations for over 15% energy savings.

04

BUSINESS IMPACT

- Identify **highest-impact end uses** linked to energy reductions
- Forecast energy + IT load to support planning scenarios
- **Produce actionable recommendations** for KPMG's Manhattan workspace
- Target: **15%+ energy savings** (efficiency + sustainability impact)



03

DATA PREPROCESSING



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**HARVARD HOUSEZERO
DATASET**

CU-BEMS DATASET

3 DATASETS

**NREL LARGE OFFICE
BUILDINGS DATASET**



HARVARD HOUSEZERO DATASET

BREAK
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TECH

Why we didn't use it

- Too sensor-heavy and not energy-focused.
 - Contains hundreds of raw sensor streams (temperatures, window states, CO₂, actuator signals, etc.) instead of the aggregated energy metrics needed for forecasting.
- Difficult to integrate.
 - Data is fragmented across many subsystems, making preprocessing and feature extraction overly complex for our workflow.
- Missing key energy-use categories.
 - Our project requires clear features like electricity total, heating total, cooling total, and plug loads—HouseZero doesn't provide these in a structured, model-ready format.

CU-BEMS DATASET

BREAK
THROUGH
TECH

Why we didn't use it

- **Climate Mismatch**
 - Data from Bangkok building (tropical climate)
 - Doesn't generalize Manhattan's seasonal heating + cooling need
- **Outdated (Pre-Pandemic)**
 - Does Not reflect hybrid work patterns or modern IT loads
- **Not Aligned with Project Scope**
 - Follows international standards
 - We prioritized North American buildings + grid assumptions (e.g., ASHRAE)



NREL LARGE OFFICE BUILDINGS DATASET

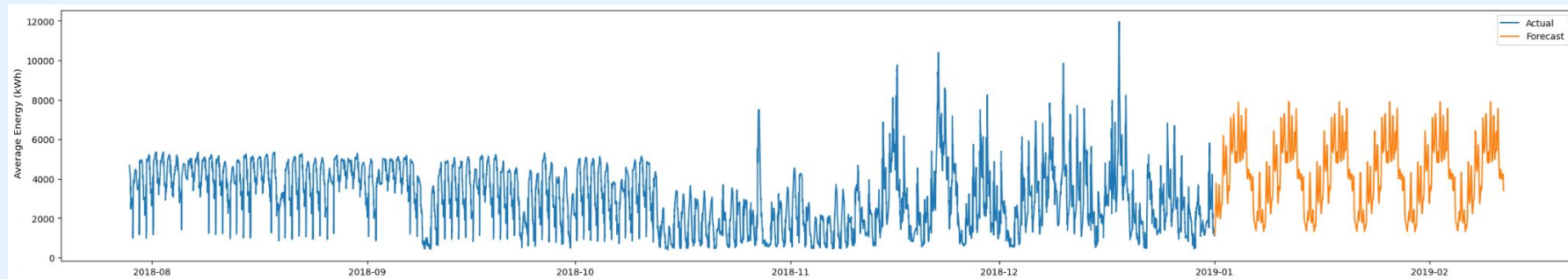
BREAK
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Why We Chose It

- Aligned with U.S. building standards
- High-resolution 15-minute interval data across a full year
- Detailed end-use breakdowns (heating, cooling, lighting, plug loads, pumps, fans, etc.)
- Includes retrofit indicators (upgrade flag + energy savings fields)
- Ideal for forecasting + AI-driven energy optimization



Deeper Data Understanding Through Forecasting



Long Short Term Memory

Seasonal Features

Increasing Energy Usage

Surface-level Gradient Boosting EDA

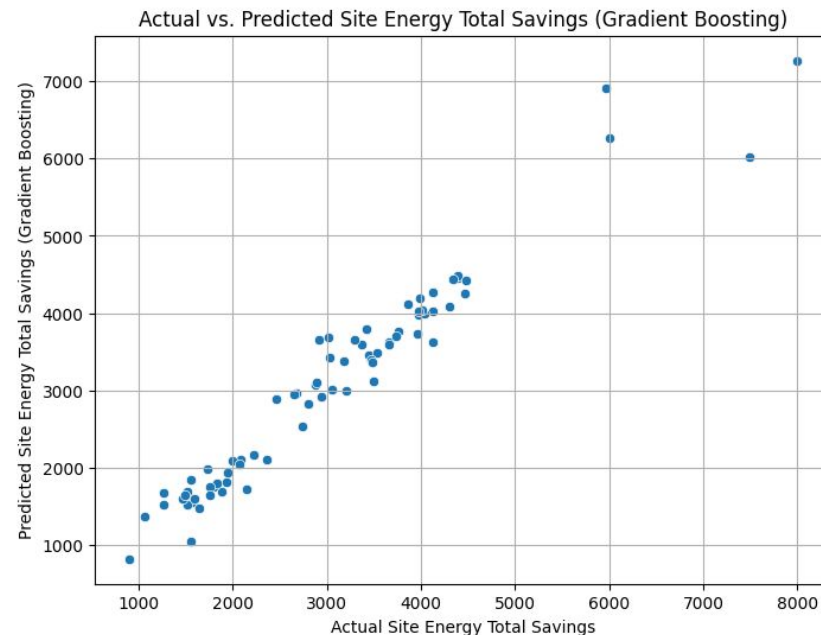
Scikit-learn

.945 R-squared

model statistically captures complexities

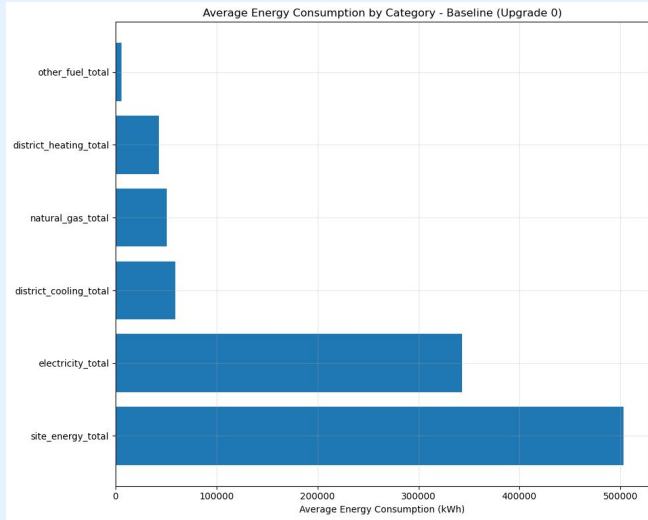
Top 6 Features:

1. electricity_heat_recovery
2. electricity_cooling
3. electricity_heating
4. electricity_pumps
5. district_heating_heating
6. electricity_fans



DATA DESCRIPTION

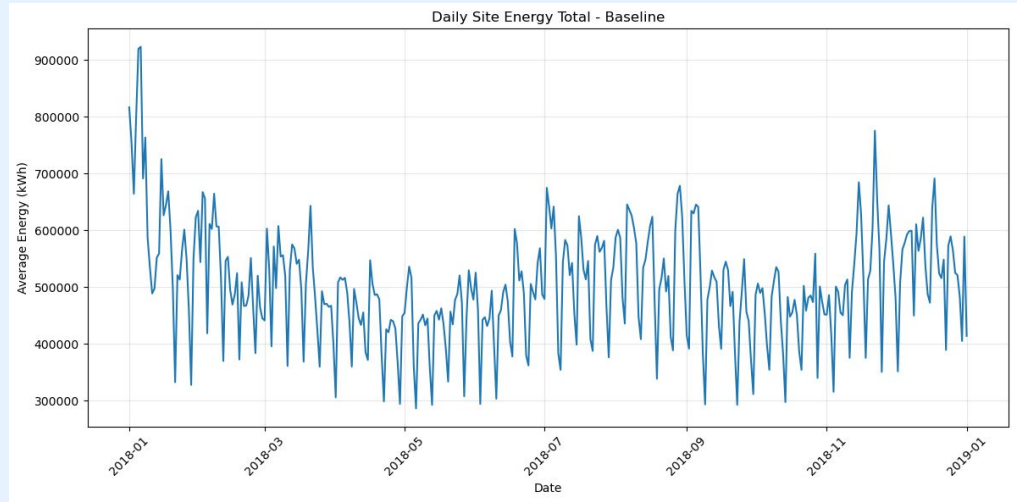
Energy Usage by Category



Baseline consumption is heavily dominated by electricity with all other categories remaining relatively small.

Electricity-driven systems dominate the building's energy profile before any upgrades.

Energy Demand by Seasons

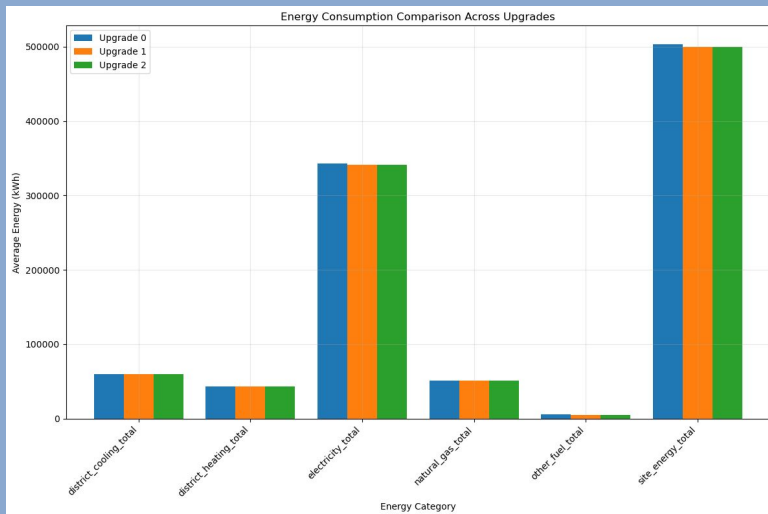


Daily energy demand fluctuates throughout the year with noticeable seasonal swings, starting high in winter, dipping in spring, and rising again toward late fall.

Seasonal heating and cooling heavily influences overall site energy use.

DATA DESCRIPTION

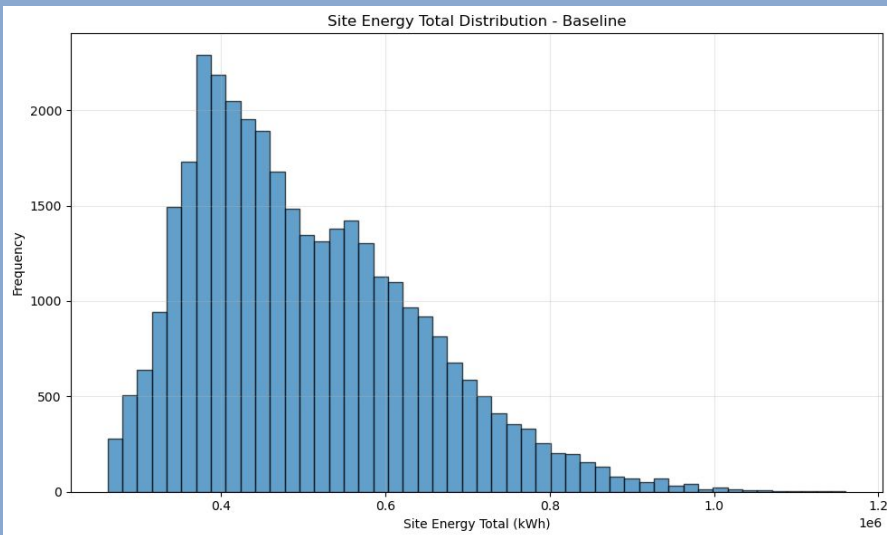
Consumption Across Updates



Energy consumption looks nearly identical across all three upgrade states, with electricity remaining the dominant load.

Upgrades 1 & 2 did not significantly change overall energy usage patterns.

Energy Total Distributions



Site energy usage clusters between 300k–600k kWh on an average.

Fairly consistent energy pattern, with occasional spikes in demand.



04

Data Understanding & Model Evaluation

Create Temporal Features

Created Temporal features
capture daily/weekly/seasonal
patterns: {hour, day_of_week,
day_name, month,
month_name, quarter,
is_weekend, season,
is_business_hours,

	timestamp	hour	day_name	month_name	season	is_weekend	is_business_hours
0	2018-01-01 00:15:00	0	Monday	January	Winter	0	0
1	2018-01-01 00:30:00	0	Monday	January	Winter	0	0
2	2018-01-01 00:45:00	0	Monday	January	Winter	0	0
3	2018-01-01 01:00:00	1	Monday	January	Winter	0	0
4	2018-01-01 01:15:00	1	Monday	January	Winter	0	0
5	2018-01-01 01:30:00	1	Monday	January	Winter	0	0
6	2018-01-01 01:45:00	1	Monday	January	Winter	0	0
7	2018-01-01 02:00:00	2	Monday	January	Winter	0	0
8	2018-01-01 02:15:00	2	Monday	January	Winter	0	0
9	2018-01-01 02:30:00	2	Monday	January	Winter	0	0

Overall Energy Consumption Statistics

Energy Breakdown by Source (Average kWh/hour):

electricity_total	:	1,365,492.92 kWh (68.3%)
natural_gas_total	:	203,150.83 kWh (10.2%)
district_heating_total	:	172,526.16 kWh (8.6%)
district_cooling_total	:	238,453.37 kWh (11.9%)

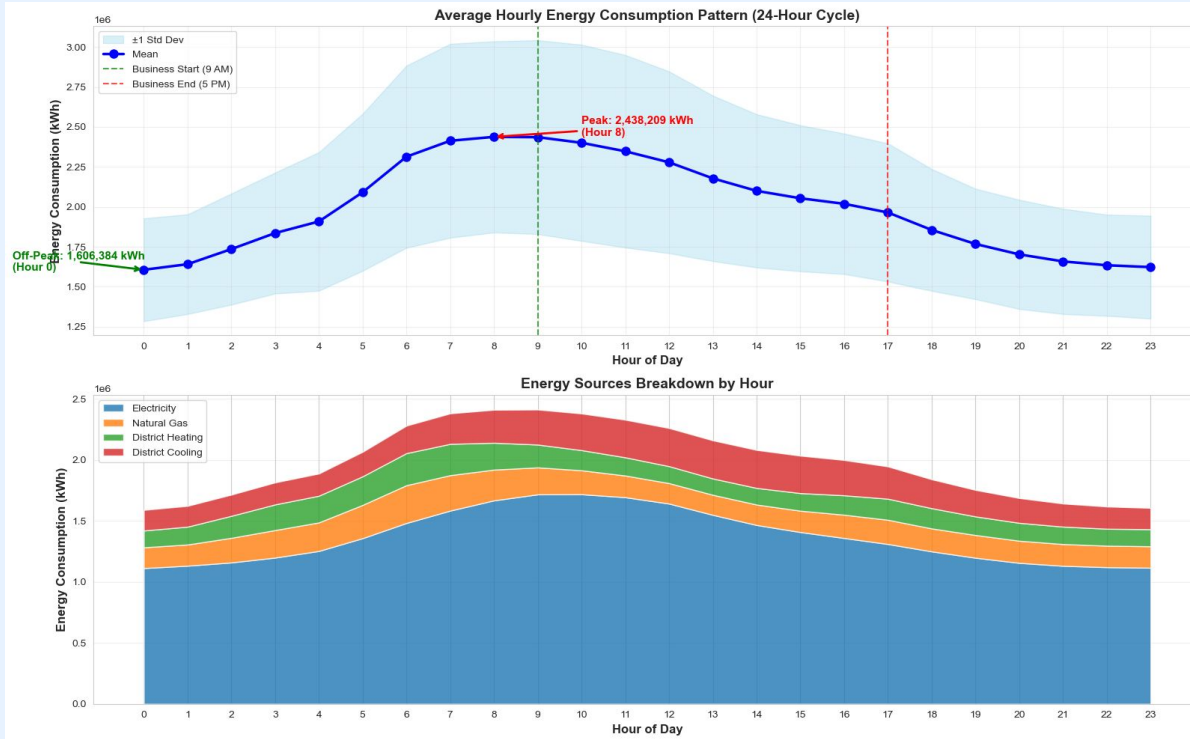
Total Site Energy (kWh):

Mean:	2,000,411.83 kWh/hour
Median:	1,899,733.35 kWh/hour
Std:	545,973.45 kWh/hour
Min:	413,187.50 kWh/hour
Max:	4,487,509.03 kWh/hour

Electricity Usage Breakdown (Average kWh/hour):

electricity_cooling	:	263,442.99 kWh (19.3%)
electricity_heating	:	42,663.15 kWh (3.1%)
electricity_interior_lighting	:	173,462.39 kWh (12.7%)
electricity_fans	:	187,921.28 kWh (13.8%)
electricity_pumps	:	38,289.54 kWh (2.8%)
electricity_interior_equipment	:	629,323.73 kWh (46.1%)

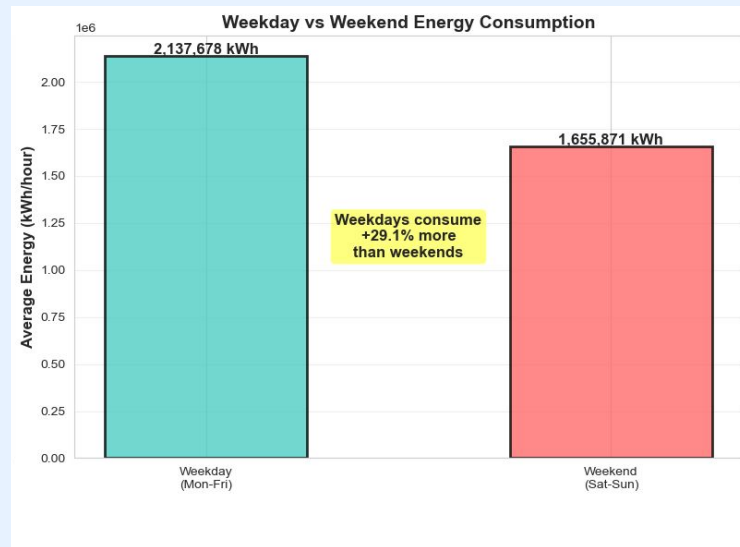
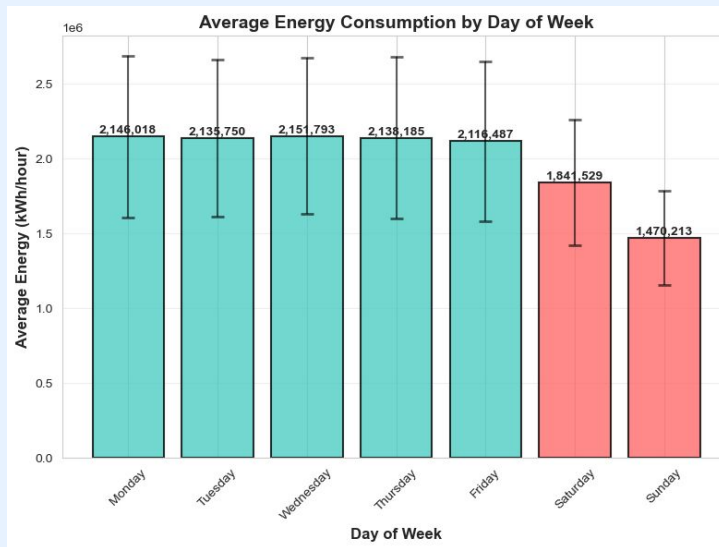
Energy by Hour of Day



HOURLY PATTERNS:

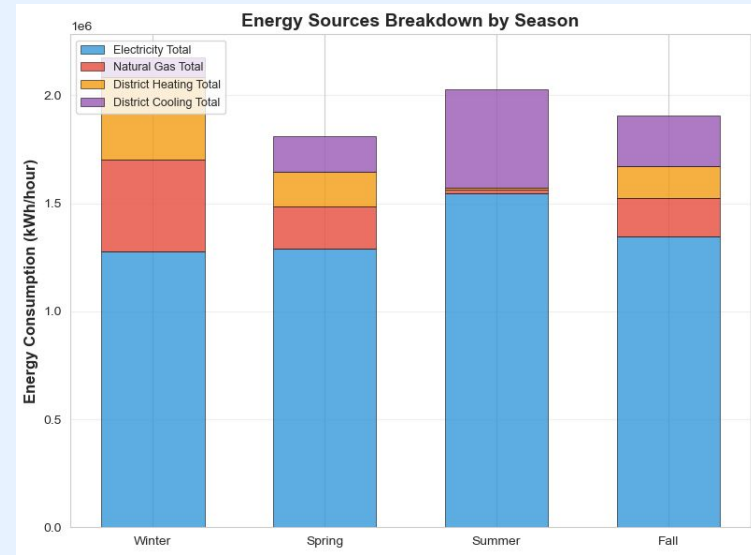
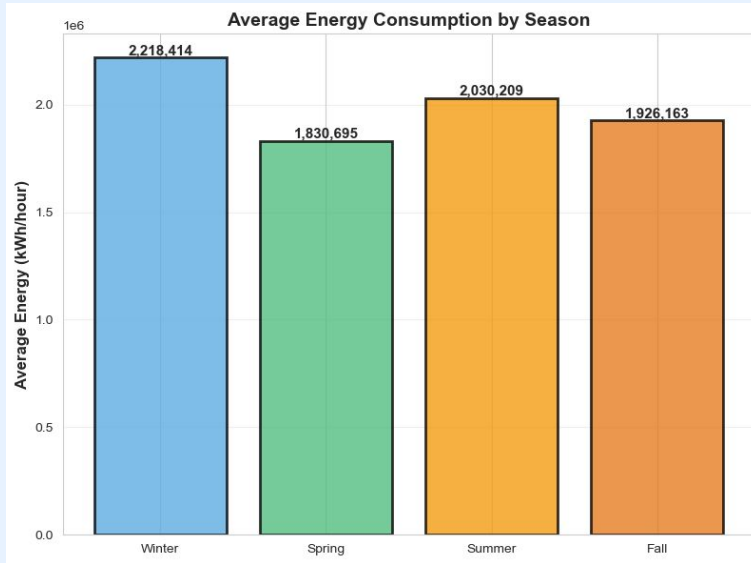
- Peak consumption: 8:00 with 2,438,209 kWh
- Lowest consumption: 0:00 with 1,606,384 kWh
- Peak is 1.5x higher than off-peak
- Business hours (9-5): ~2,227,068 kWh/hour
- After hours: ~1,887,151 kWh/hour

Energy by Day of Week: Weekday vs Weekend



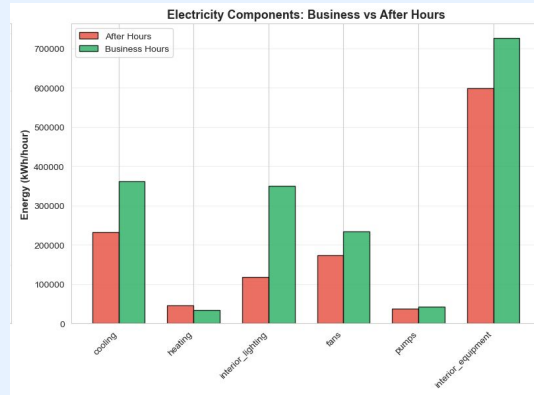
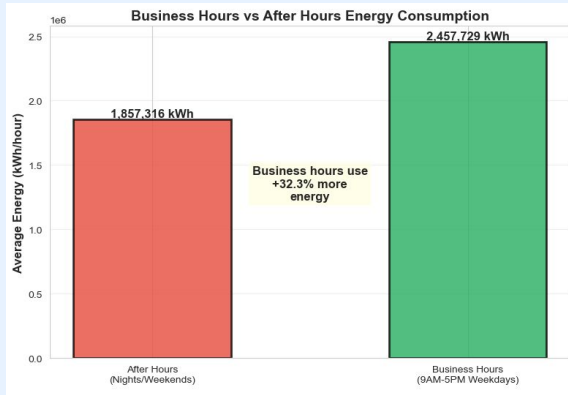
- Weekday average: 2,137,678 kWh/hour
- Weekend average: 1,655,871 kWh/hour
- Weekdays use +29.1% more energy than weekends

Energy by Season



- Peak seasons: Winter & Summer
- Low season: Spring
- Seasonal variation: 19.4%

Business Hours vs After Hours



- Business hours avg: 2,457,729 kWh/hour
 - After hours avg: 1,857,316 kWh/hour
 - Difference: +32.3%
- Hours in business: 2,088 hours (23.8%)
- Hours after/weekend: 6,673 hours (76.2%)

Preparing Data for Modeling

- Temporal features: Hour, day, month, business hours flags
- Lag features: Previous 1h, 24h, 168h (1 week) consumption
- Rolling averages: 24h and 168h moving averages

Feature list:

1. hour
 2. day_of_week
 3. month
 4. is_weekend
 5. is_business_hours
 6. energy_lag_1h
 7. energy_lag_24h
 8. energy_lag_168h
 9. energy_rolling_24h
 10. energy_rolling_168h
-

Baseline: 24-hour moving average

Baseline Performance Analysis:

- Accuracy 32.5% (considered accurate within $\pm 10\%$ range!)
- R^2 : 0.2767 – Explains only 27.7% of variance
- MAPE: 17.06% – Average error is 17.1% of actual value
 - Simple moving averages struggle with energy pattern

Why Baseline is Weak:

- Cannot capture complex temporal patterns
 - Misses weather-driven demand changes
 - No understanding of operational schedules
 - Limited to recent historical averaging
-

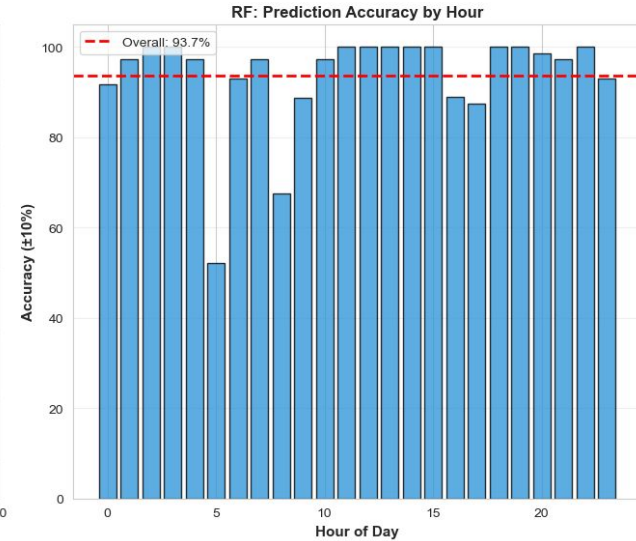
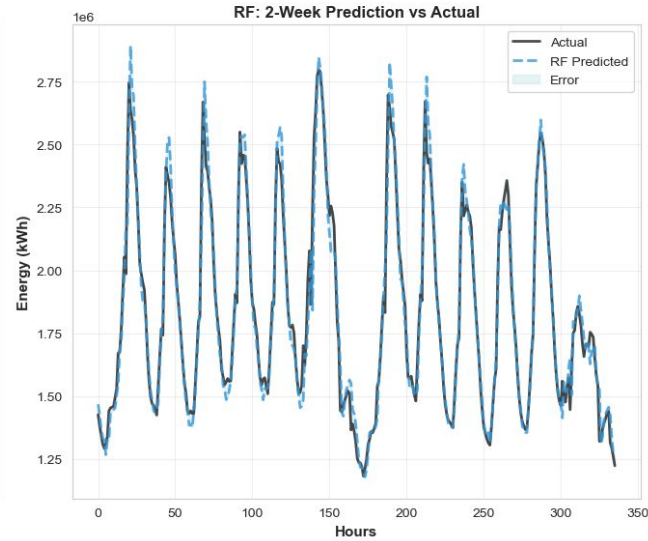
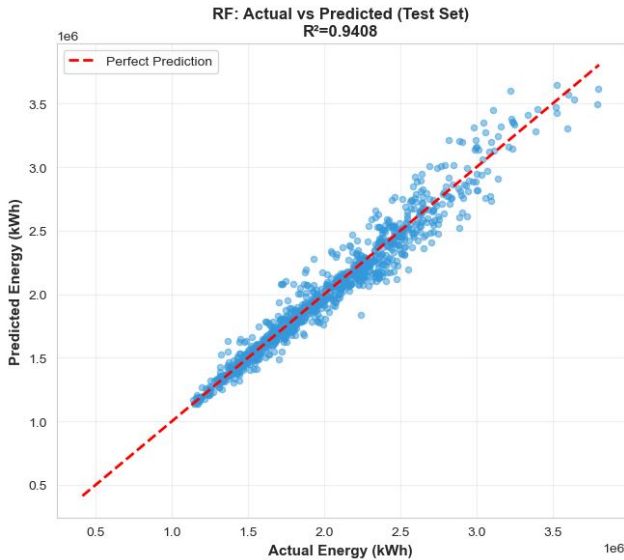
Random Forest

- It handles non-linear relationships exceptionally well → energy consumption doesn't follow straight lines
- It's robust to outliers and missing data, which is critical for real-world deployment
- It provides feature importance rankings, so we know which factors drive consumption"

Top 5 Most Important Features:

energy_lag_1h	: 0.9493
hour	: 0.0299
energy_lag_168h	: 0.0046
energy_rolling_24h	: 0.0036
energy_lag_24h	: 0.0033

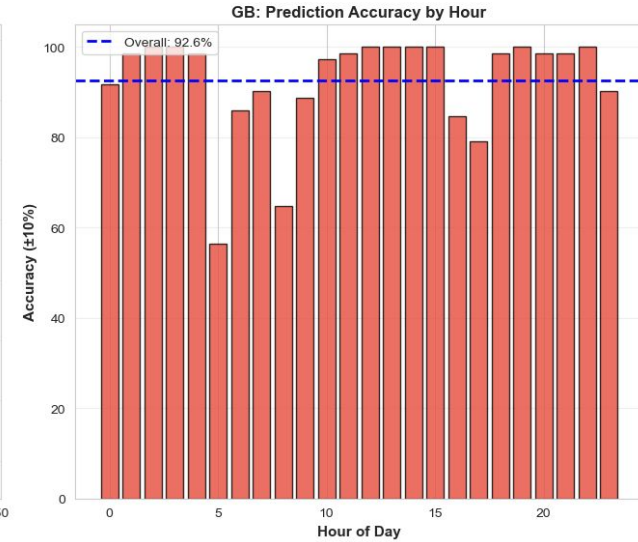
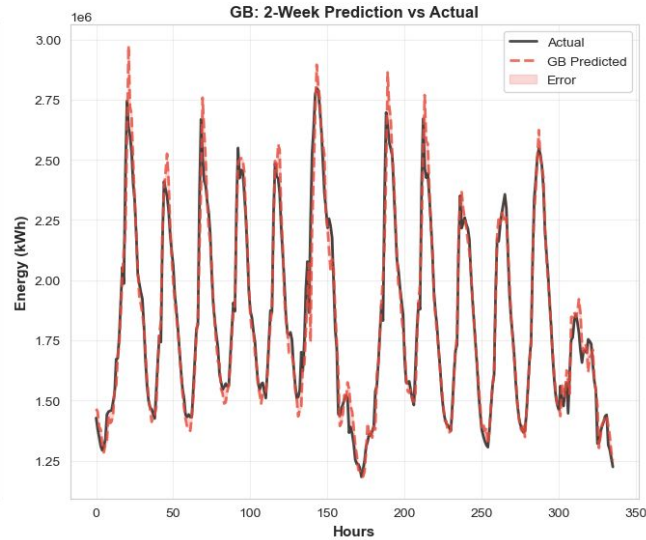
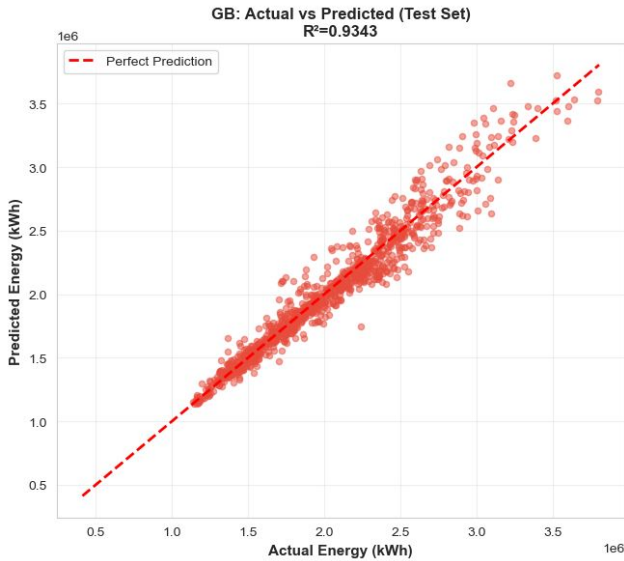
- accuracy : 93.7% – almost 94 out of 100 predictions are within $\pm 10\%$ of actual energy consumption
- R^2 : 0.94 – explains 94% of energy variance
- MAPE: 3.87% – average error under 4%



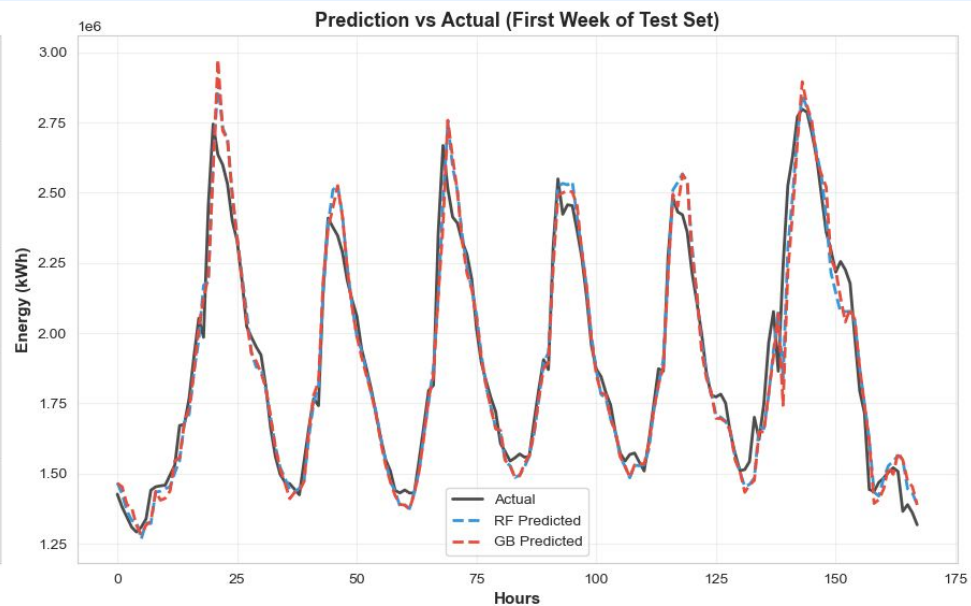
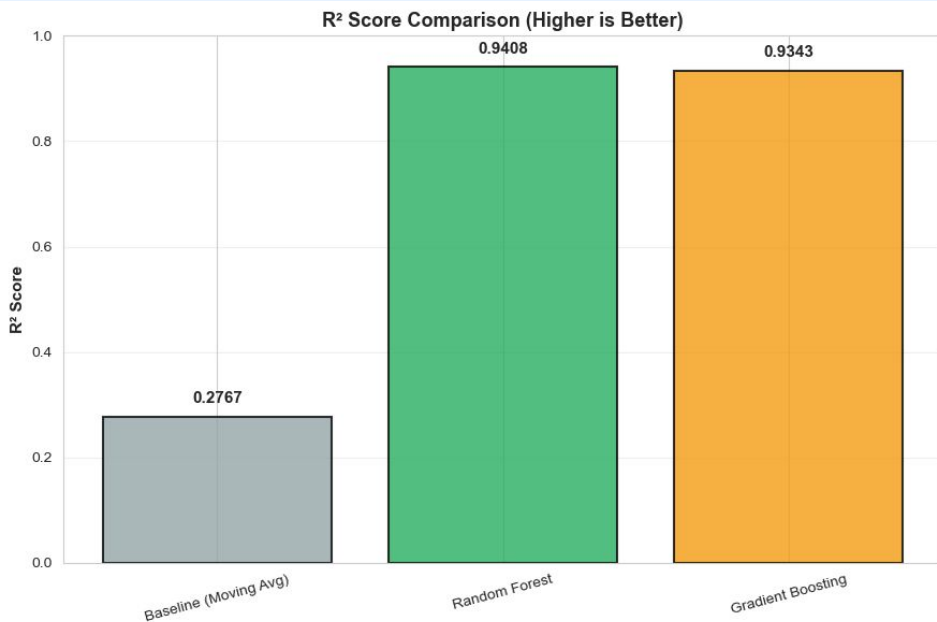
Gradient Boosting

- It achieves the highest precision on complex patterns, perfect for capturing subtle daily/seasonal variations
 - It excels at learning from residual errors, making it ideal for edge cases
 - Industry-standard for time-series forecasting (used by Uber, Airbnb for demand prediction)
-

- Accuracy: 92.6
- $R^2 = 0.93$ – explains 93% of variance
- MAPE = 4.02 – average error of 4%

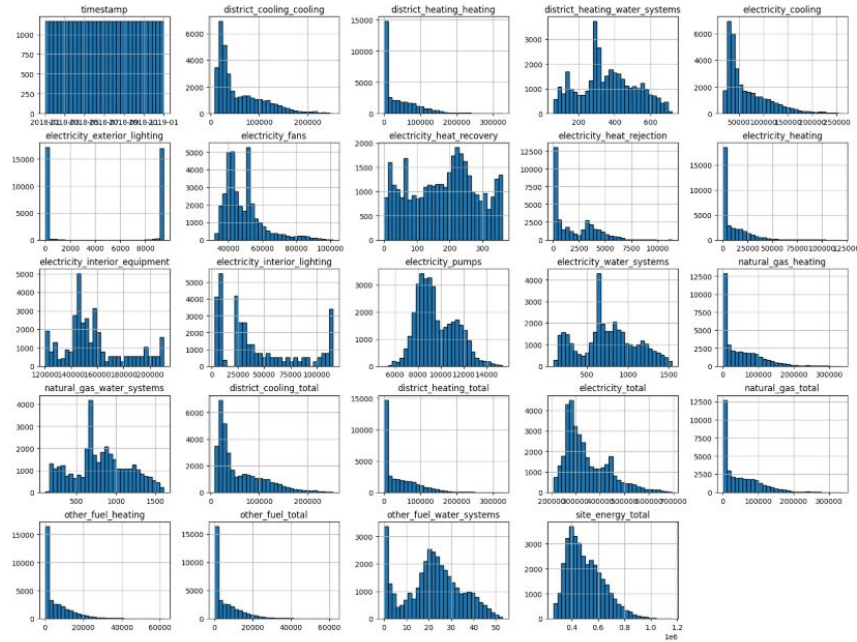


Model Comparisons



Feature Selection

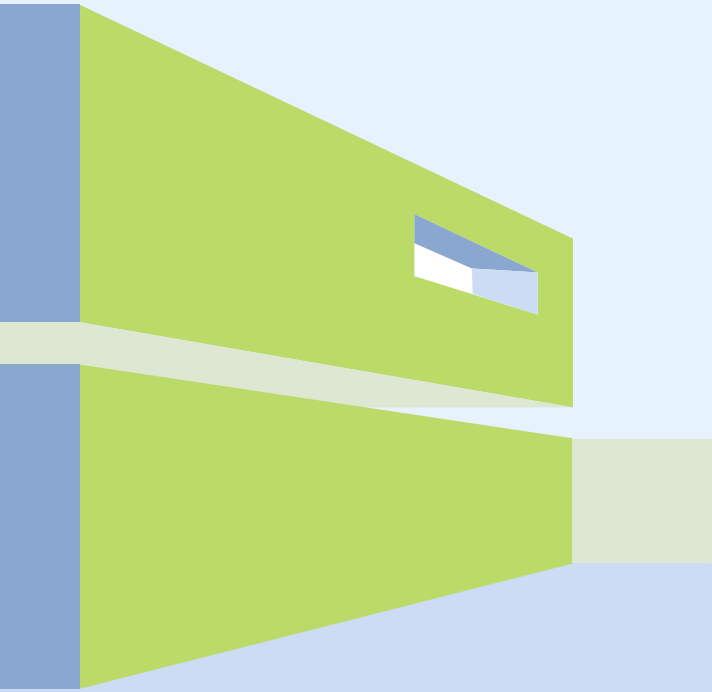
Feature Distributions



Explain why we used these graphs and what it shows us in bullet points

Model Comparison

Model Name	Description	Results	Pros	Cons
Simple Linear Regression	Using sklearn library, straight forward regression	$\sim .8 r^2$	straightforward, can see which variables are most influential	does not capture complexity and relationships between variables
Ordinary Least Squares	Using statsmodels library	0.923 R-squared	provides significantly more evaluation metrics	Integration with ML workflow
Gradient Boosted Regression	Sklearn, straightforward to use	.945 R-squared	model is more descriptive and captures complexities	results could be due to overfitting

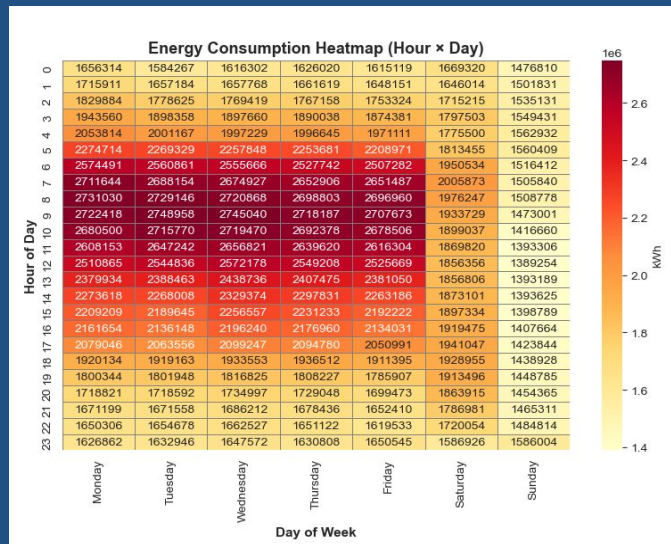


05

Conclusions, Feedback for *KPMG*, and Next Steps

Overall Findings

- High average energy use: about 2.0M kWh consumed per hour on average
- Clear energy pattern usage $\Rightarrow$ accurate forecasting
 - Work schedule drives demand: business hours are ~32% higher than after-hours.
 - Energy peaks around 8:00 AM (~2.44M kWh) vs lowest around 12:00 AM (~1.61M kWh), meaning peak load is ~1.5 $\times$ off-peak.
 - Weekdays use ~29% more energy than weekends.
 - Year seasonal variation with highest demand in winter and summer
- Main drivers: **electricity and heating/cooling-related usage** have the biggest impact on predicting energy consumption.

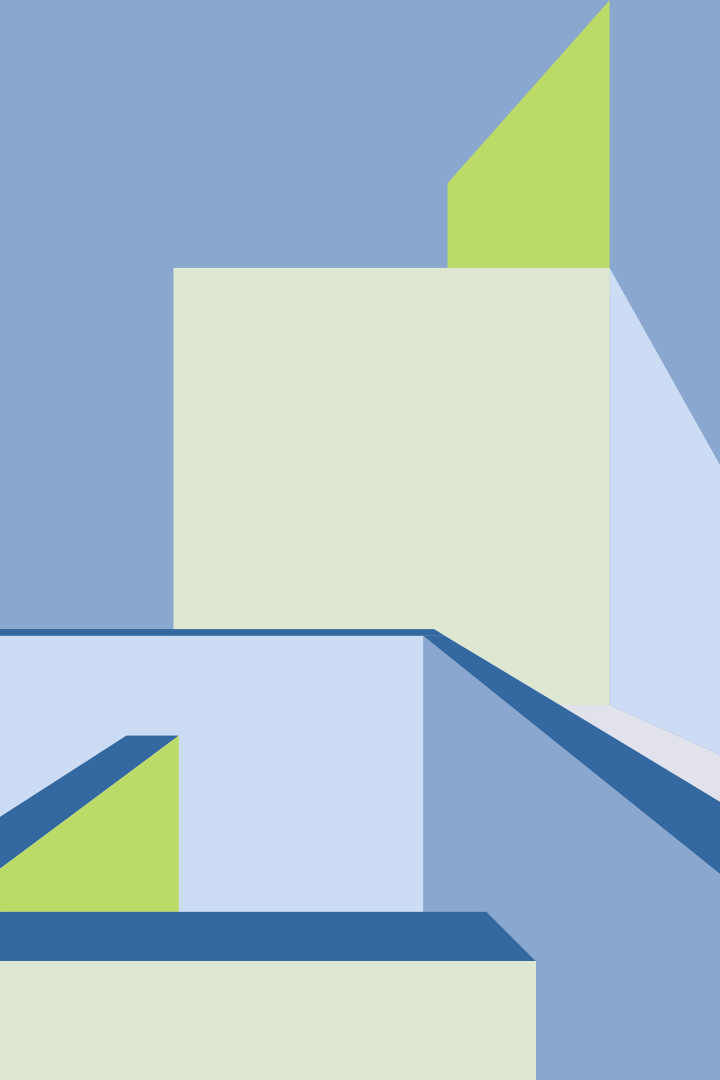


Feedback for *KPMG*

Based on our findings we recommend:

- Electricity has a large impact on energy usage ⇒ Install **sensor lighting, smart controls, smart plugs** in conference rooms, restrooms, and corridors.
- Heating and cooling has large impact on energy usage ⇒ Install **programmable thermostats** with time-of-use optimization
- Set HVAC to 'setback mode' outside 7AM-7PM weekdays





Potential Next Steps:

Model upgrades: Tune Random Forest, try XGBoost/LightGBM, and add simple baseline models for comparison and explainability.

Run Scenarios: Run “what-if” scenarios (ex: reduce after-hours load by X%, shift peak load, adjust HVAC schedules) and estimate impact.

Build a dashboard: create a simple dashboard for hourly trends, peak alerts, weekday/weekend splits, and forecasted usage.

Questions?

